Hardy-Rogers (Ćirić-type) Contractions

HARDY-ROGERS CONTRACTION (1973)

Definition: A mapping $T: X \to X$ is called a Hardy-Rogers contraction if there exist nonnegative numbers $a_1, a_2, a_3, a_4, a_5 \in [0, 1)$ such that

$$a_1 + a_2 + a_3 + a_4 + a_5 < 1$$

and the following condition holds:

$$d(Tx, Ty) \leq a_1 d(x,Tx) + a_2 d(y,Ty) + a_3 d(x,Ty) + a_4 d(y,Tx) + a_5 d(x,y)$$

for all $x, y \in X$.

KEY INSIGHT: This is a generalization of the Ćirić-type contractions, combining:

- Distance terms involving the mapping ($d(x,Tx)$, $d(y,Ty)$)
- Distance terms mixing points and images ($d(x,Ty)$, $d(y,Tx)$)
- Direct distance between points ($d(x,y)$)

THEOREM: If (X, d) is a complete metric space and T is a Hardy-Rogers contraction, then T has a unique fixed point $u \in X$, and for any $x \in X$, the sequence $\{T^n x\}$ converges to u.

SPECIAL CASES:

- Set $a_3 = a_4 = 1/2$, others = 0 → Zamfirescu contraction
- Set $a_1 = a_2 = r$, others = 0 → Kannan contraction
- Set $a_5 = k$, others = 0 → Banach contraction